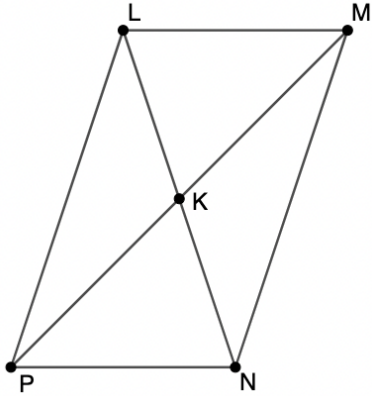


Geometry
Unit 8
Lesson 1 Practice

Name _____
Date _____
Period _____

Parallelogram $LMNP$ where diagonals $\overline{LN}$ and $\overline{MP}$ intersecting at K as shown.



Given: $LMNP$ is a parallelogram with diagonals $\overline{LN}$ and $\overline{MP}$ intersecting at K .

Prove: $\angle LMN \cong \angle NPL$

Complete the following two-column proof.

Statement	Reason
$LMNP$ is a parallelogram with diagonals $\overline{LN}$ and $\overline{MP}$	Given
$\overline{LM} \cong \overline{PN}; \overline{LP} \cong \overline{MN}$	1.
2.	Reflexive Property of Congruence
3.	4.
$\angle LMN \cong \angle NPL$	5.

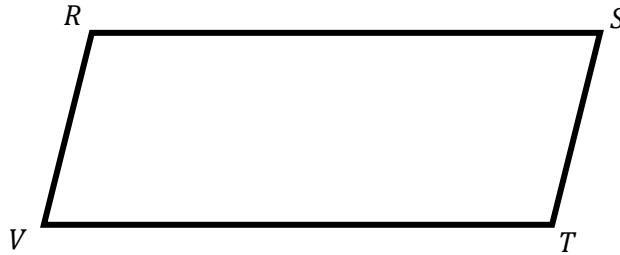
Fill in the blanks with the correct terms to complete the statements.

6. Consecutive interior angles of a parallelogram are _____.
7. The diagonals of a parallelogram _____ each other.
8. Opposite angles of a parallelogram are _____.
9. Opposite _____ of a parallelogram are parallel.
10. A quadrilateral containing _____ set(s) of congruent sides is a
_____.

Geometry
Unit 8
Lesson 2 Practice

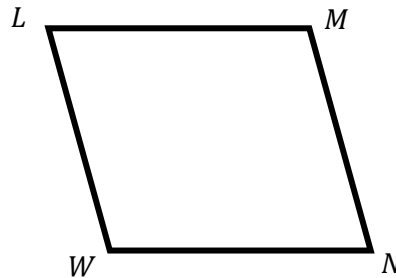
Name _____
Date _____
Period _____

Parallelogram $RSTV$ is shown where $m\angle S = (4x - 8)^\circ$, $m\angle T = 144^\circ$, and $m\angle V = 36^\circ$.



1. I can conclude ...
2. I know this because ...
3. Since $RSTV$ is a parallelogram, what is the value of x ?

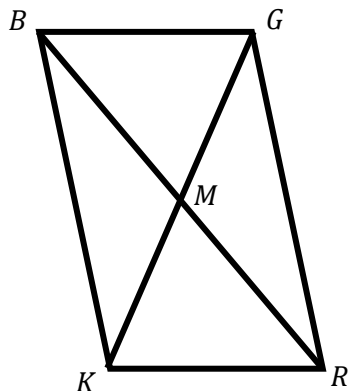
Quadrilateral $LMNW$ is shown where $m\angle M = (14x - 8)^\circ$, $m\angle N = (16y - 4)^\circ$, $m\angle W = (5x + 64)^\circ$, and $m\angle L = (12y + 16)^\circ$.



4. What value of x will make $LMNW$ a parallelogram?
5. What value of y will make $LMNW$ a parallelogram?
6. Given that $LMNW$ is a parallelogram, what is the $m\angle M$?

Quadrilateral $BGRK$ is shown with diagonals $\overline{BR}$ and $\overline{GK}$ that intersect at point M .

$MB = 8z - 6$, $MR = 32$, $MG = 28$, and $MK = 3t + 3$, where $t = 8$.



Charlene and Jordan are discussing whether $BGRK$ is a parallelogram. Charlene says $BGRK$ is a parallelogram and Jordan says $BGRK$ is not a parallelogram.

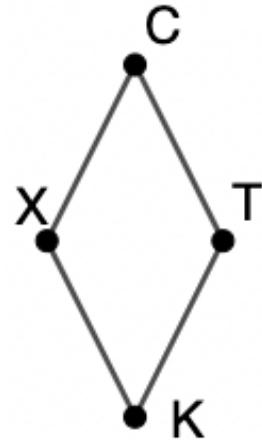
7. Who is correct?
8. Which property of parallelograms was used to determine if $BGRK$ is a parallelogram?
9. Given that $BGRK$ is a parallelogram, what is the value of z ?
10. What is the length of $\overline{MK}$?

Geometry
Unit 8
Lesson 3 Practice

Name _____
Date _____
Period _____

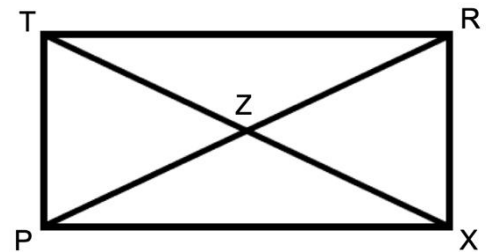
Rhombus $CTKX$ is shown where $CT = 15x - 3$ and $TK = 7x + 1$.

1. Select all true properties of $CTKX$.
 - ☐ Opposite sides are parallel.
 - ☐ Opposite sides are congruent.
 - ☐ Consecutive angles are congruent.
 - ☐ All angles are right angles.
 - ☐ Diagonals are perpendicular.
 - ☐ All sides are congruent.
2. What is the value of x ?
3. What is the length of $\overline{CT}$?



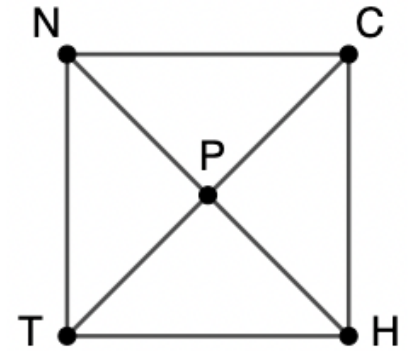
Rectangle $TRXP$ is shown with diagonals $\overline{TX}$ and $\overline{RP}$ which intersect at point Z .

4. Write two congruent statements using the diagonals of $TRXP$.



5. $TZ = (5y - 8)$, and $ZX = (3y + 16)$. What is the value of y ?
6. What is the length of $\overline{RP}$?

Quadrilateral $NCHT$ is shown with diagonals $\overline{NH}$ and $\overline{CT}$ intersecting at point P .

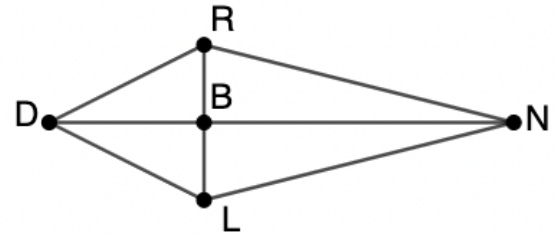


7. What value of x makes $NCHT$ a square, if $m\angle TNC = 90^\circ$, $m\angle NCH = 90^\circ$, $CH = 21$, and $HT = (4x - 7)$?
8. Given $NCHT$ is a square with $PC = (5y - 2)$ and $PT = (3y + 4.74)$. What is the length of $\overline{NH}$, ?
9. What are some similarities between $CTKX$ and $NCHT$?
10. What are some differences between $TRXP$ and $NCHT$?

Geometry
Unit 8
Lesson 4 Practice

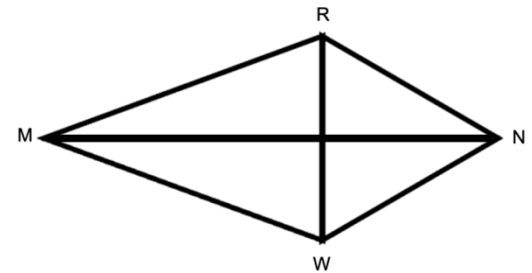
Name _____
Date _____
Period _____

Kite $RNLD$ is shown with diagonals $\overline{ND}$ and $\overline{RL}$ intersecting at point B where $BD = 24$ and $RB = 15$.



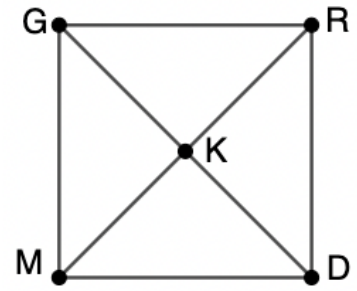
1. What is the length of $\overline{RD}$?
2. Given $NB = 12$. What is the length of $\overline{NR}$?
3. What is the perimeter of $RNLD$?

$MRNW$ is a kite where $m\angle RMN = (5x - 17)^\circ$, $m\angle MRN = (11x + 13)^\circ$, $m\angle WNM = (3y - 2)^\circ$, and $m\angle MWN = 112^\circ$.



4. What is the value of x ? Round your answer to the nearest hundredth.
5. What is the value of y ? Round your answer to the nearest hundredth.
6. What is the measure of $\angle RNW$?

Quadrilateral $GRDM$ is shown with diagonals $\overline{GD}$ and $\overline{MR}$ intersecting at point K .



7. What must be true for $GRDM$ to be a rectangle?

8. Given $DK = 9z$, and $DM = 18$. What value of z will make $GRDM$ a square?

9. Given $GRDM$ is a square, what is the length of $\overline{RM}$?

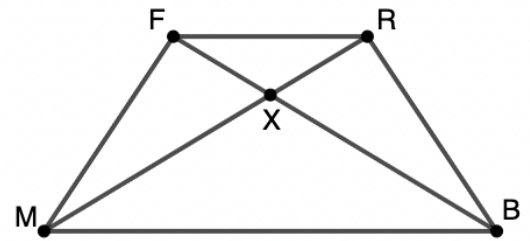
10. Given $GRDM$ is a square, what is the length of $\overline{GM}$?

Geometry
Unit 8
Lesson 5 Practice

Name _____
Date _____
Period _____

Given: $FRBM$ is an isosceles trapezoid, $\angle MFX \cong \angle BRX$.

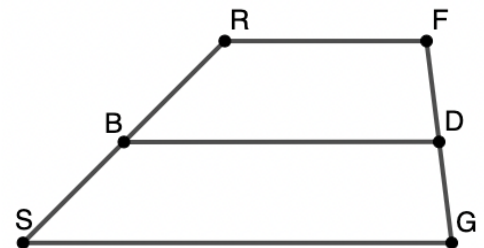
Prove: $\triangle MXB$ is an isosceles triangle.



Statement	Reason
$FRBM$ is an isosceles trapezoid	Given
$\angle FMB \cong \angle RBM$	1.
$\angle MFX \cong \angle BRX$	Given
2.	Reflexive Property of Congruence
3.	AAS Congruence
4.	Corresponding Parts of Congruent Triangles are Congruent
$\triangle MBX$ is an isosceles triangle	5.

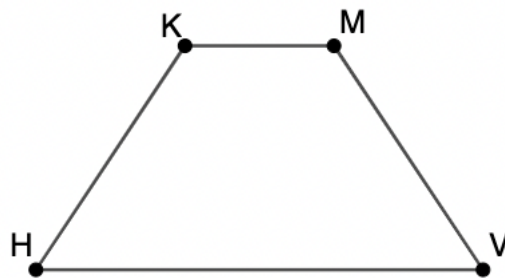
Trapezoid $SRFG$ is shown. $\overline{BD}$ is a midsegment of $SRFG$.
 $FD = 4x + 34$, $DG = 8x - 14$, $RB = 8y + 41$, $SB = 13y - 4$.

6. Find the values of x and y .



7. Find the lengths of $\overline{RS}$ and $\overline{DG}$.

Isosceles trapezoid $KMVH$ is shown where $m\angle KHV = (16z - 6)^\circ$, $m\angle HKM = (21z + 38)^\circ$, $m\angle KMV = (16t - 26)^\circ$.



8. What is the value of z ?

9. What is the value of t ?

10. What are $m\angle VHK$, $m\angle HKM$, $m\angle KMV$, and $m\angle MVH$?

Geometry
Unit 8
Lesson 6 Practice

Name _____
Date _____
Period _____

Two congruent angles of an isosceles trapezoid have measures $(8x + 14)^\circ$ and $(12x - 8)^\circ$.

1. What is the value of x ?

2. What are the measures of each angle in the trapezoid?

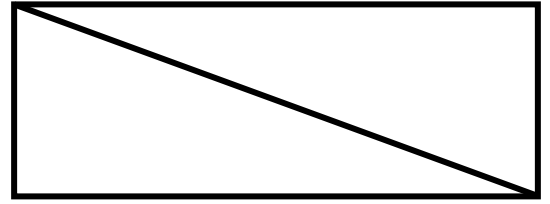
For each quadrilateral description, select all shapes that match the description.

3. Quadrilateral $GRMW$ has two pairs of congruent sides and consecutive angles are supplementary.
 - ☐ Parallelogram
 - ☐ Rectangle
 - ☐ Square
 - ☐ Rhombus
 - ☐ Kite

4. Quadrilateral $BNTV$ has four congruent sides and diagonals are perpendicular to each other.
 - ☐ Parallelogram
 - ☐ Rectangle
 - ☐ Square
 - ☐ Rhombus
 - ☐ Kite

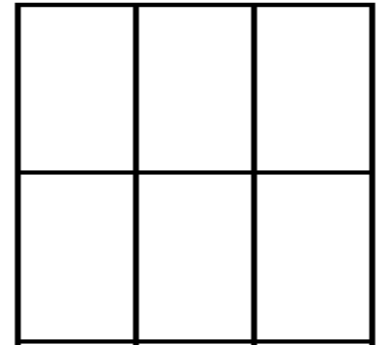
5. Quadrilateral $KPSX$ has two pairs of congruent sides and one pair of opposite congruent angles.
 - ☐ Parallelogram
 - ☐ Rectangle
 - ☐ Square
 - ☐ Rhombus
 - ☐ Kite

Barrett is looking at a building on the school grounds which is shaped like a rectangle as shown. He knows the building has a height of $32x$ feet, width of $46x$ feet and the diagonal of the building is 200 feet.



6. What is the value of x ? Round answers to the nearest hundredth.
7. What is the height of the building? Round answers to the nearest hundredth.
8. What is the width of the building? Round answers to the nearest hundredth.

A wall paneling with a total area of 18 square feet is made up of 6 rectangles, as shown. There are two congruent rectangles along the height and three across the width. Each rectangle in the paneling is $(3x + 1)$ feet in height and $(2x - 1)$ feet in width.

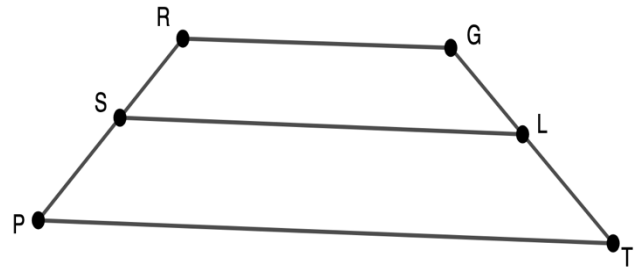


9. What is the value of x ? Round your answer to the nearest hundredth.
10. What are the dimensions of the portion of the wall covered by the paneling?

Geometry
Unit 8
Lesson 7 Practice

Name _____
Date _____
Period _____

$PTGR$ is a trapezoid as shown where $\overline{PT} \parallel \overline{RG}$. $\overline{SL}$ is the midsegment of $PTGR$ with $SL = 18$, $PT = 9x - 5$, and $RG = 7x - 3$.



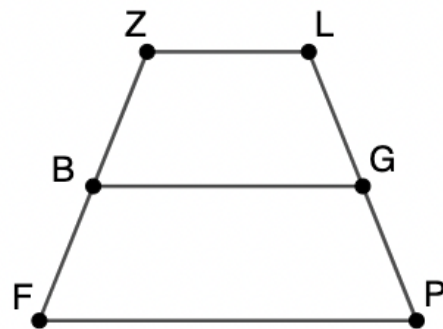
- Write an equation to show the relationship between $\overline{SL}$ and the bases of the trapezoid.
- What is the value of x ? Round your answer to the nearest hundredth.
- What are the lengths of $\overline{PT}$ and $\overline{RG}$? Round your answers to the nearest hundredth.

Trapezoid $HKMV$ is shown with midsegment $\overline{DP}$ where $VH = 9x + 12$, $MK = 6x$, and $DP = 10x$.



- What is the value of x ? Round your answer to the nearest tenth.
- What is the length of $\overline{DP}$?
- What are the lengths of $\overline{VH}$ and $\overline{MK}$? Round your answer to the nearest tenth.

Isosceles trapezoid $FPLZ$ is shown with midsegment $\overline{BG}$ where $ZL = 5x + 1$, $FP = 8x + 2$, $BG = 4x + 4$, $BZ = 3.5$.



7. What is the value of x ?

8. What is the length of $\overline{BG}$?

9. What are the lengths of $\overline{ZL}$ and $\overline{FP}$?

10. What is the perimeter of $FPLZ$?